ET AI HACKATHON: AGENTIC EDITION · PROBLEM STATEMENT 1

regComply-AI

Agentic Banking Regulatory Intelligence Platform

A 7-agent AI pipeline that transforms raw regulatory documents into an actionable, traceable compliance intelligence chain — from obligation extraction to board-level remediation plans.

11
of 14 features covered

7
specialized AI agents

F3/D2
9-blocker position

Banking · RegTech | Agentic AI | Full-Stack Prototype

MEET THE TEAM

Team Introduction

RS

Rahul Sain
Co-developer

AB

Ansh Bhardwaj
Co-developer

WHAT WE BUILT

- ✓ Solution architecture & agentic pipeline design
- ✓ Backend engineering (FastAPI, Python)
- ✓ LLM prompt engineering (OpenAI / Groq / Anthropic)
- ✓ Frontend UI / UX (SSE streaming dashboard)
- ✓ Domain specialization: Indian banking regulation (RBI, SEBI, Basel III)

AI/ML RegTech Full-Stack

SUBMISSION CONTEXT

Problem Statement

> *"Develop an agentic solution that understands regulatory requirements and how they change over time, assesses banks' internal control mechanisms, identifies gaps, and suggests appropriate remedies."*

Why this problem?

- ✓ Banks paid ~$321B in fines between 2008 and 2016 (BCG Global Risk 2017)
- ✓ Manual gap analysis is slow and document-heavy; our pipeline does a first pass in one automated run
- ✓ Regulations change constantly — change intelligence is table stakes
- ✓ Perfect match for agentic AI: multi-step reasoning, traceable outputs

THE CHALLENGE

Banking Compliance Is Broken

~$321B

in fines paid by banks between 2008 and 2016 (BCG Global Risk 2017)

Manual

obligation-to-control mapping usually lives in spreadsheets maintained by analysts

Reactive

gaps tend to surface at audit or inspection time, not before

Current State Pain Points

- ✓ Regulations ingested manually — spreadsheets, emails, sticky notes
- ✓ No traceability: can't link a control to the specific obligation it addresses
- ✓ Gap discovery is reactive — found during audits, not proactively
- ✓ Regulatory changes require expensive re-analysis from scratch
- ✓ Remediation plans are ad hoc with no risk prioritization
- ✓ No "what-if" capability to model new regulatory scenarios

The Core Challenge

Regulations → Obligations → Controls → Evidence → Gaps → Remediation → Monitoring

Maintaining this continuous, evolving, and auditable chain is the core problem — and it is exactly the chain regComply-AI's agents build.

OUR SOLUTION

regComply-AI: The Intelligent Compliance Chain

Banks upload regulations and internal control policies. Seven specialized agents analyze, map, score and remediate, producing a traceable compliance picture.

What makes it agentic

- ✓ Each agent has a **single, specialized role**; its output feeds the next agent
- ✓ Agents **reason over structured JSON** with confidence scoring
- ✓ JSON recovery logic handles partial/malformed LLM outputs
- ✓ Pipeline streams real-time progress via Server-Sent Events
- ✓ Simulation agent reasons over hypothetical future scenarios

Compliance score (deterministic, computed from agent mappings)

```
score = (Full + 0.5 × Partial) ÷ Total × 100
```

Solution Output: one compliance intelligence report

- ✓ Obligations with regulation references
- ✓ AI-scored controls (0–100)
- ✓ Full / Partial / None mapping + confidence
- ✓ Risk-ranked gap register
- ✓ Remediation: owner, timeline, effort
- ✓ Board-level executive summary
- ✓ Regulatory change impact analysis
- ✓ What-if score projections

Three operating modes

ANALYZE Full 6-agent pipeline: obligations → controls → mapping → gaps → remediation

CHANGE Regulatory change intelligence: compare old vs. new regulation versions

SIMULATE What-if scenarios: model proposed regulations before they take effect

TECHNICAL ARCHITECTURE

7-Agent Sequential Pipeline

INPUT LAYER
PDF · DOCX · TXT · Plain text paste

OUTPUT LAYER
SSE stream · JSON export · UI dashboard

Agent 1
Obligation Extractor
Extracts 6–8 structured obligations per regulation with categories & references

→

Agent 2
Control Analyzer
Extracts 5–8 controls with effectiveness score (0–100), type & automation level

→

Agent 3
Coverage Mapper
Maps obligations → controls: Full / Partial / None with confidence score

→

Agent 4
Gap Analyzer
Risk-scores gaps 1–100 (Critical / High / Medium / Low) with reasoning

→

Agent 5
Remediation
Action plans with owner, timeline, effort + board-level executive summary

Agent 6 — Regulatory Change Intelligence
Diffs old vs. new regulation → detects New / Amended / Removed / Stricter / Relaxed requirements with CRO impact summary

Agent 7 — What-if Simulation
Takes natural-language scenario → projects compliance score delta, new gaps, required actions, simulation narrative

INFRASTRUCTURE: FastAPI + uvicorn · SSE streaming · OpenAI / Groq / Anthropic (switchable) · Vanilla JS dashboard · pdfplumber + python-docx · Pydantic v2 schemas

AI MODELS & TECHNOLOGIES

Engineering Stack

LLM Layer · one env var switches provider

OpenAI (used in the demo)
gpt-4.1 · temperature 0.1 · model set via OPENAI_MODEL

Groq
openai/gpt-oss-120b · OpenAI-compatible endpoint · temperature 0.1

Anthropic Claude (Alternative)
claude-haiku-4-5-20251001 · same JSON-schema prompts

Reliability Engineering

- ✓ Markdown fence stripping from LLM output
- ✓ Thinking tag removal (Qwen3-style <think> blocks)
- ✓ Truncated JSON array recovery
- ✓ Depth-tracking JSON salvage on parse failure

Full Tech Stack

Python 3.10+
Primary language

FastAPI
REST + SSE API

uvicorn (ASGI)
Async server

pdfplumber
PDF extraction

python-docx
Word extraction

Pydantic v2
Schema validation

SSE (EventSource)
Real-time streaming

Vanilla JS/HTML
No-framework UI

Prompt Strategy

All agents use **schema-constrained JSON prompting**, with no orchestration framework (LangChain, AutoGen, CrewAI). Conservative instruction: *"Gaps are valuable findings — do not force-fit poor matches."*

PRODUCT DEMO

Walkthrough: RBI KYC Compliance Analysis

1. **Upload Documents**
   RBI Master Direction on KYC (regulation) + Bank's KYC Policy Manual (controls). Drag & drop or paste text.

2. **Live Pipeline Execution**
   SSE stream shows each agent activating in real time — obligation extraction → control scoring → mapping → gap analysis → remediation.

3. **Compliance Dashboard**
   Score gauge, coverage stat cards, executive summary and a 5-tab report: Traceability · Gaps · Remediation · Controls · Obligations.

4. **Change Intelligence**
   Compare the 2022 and 2024 versions. Agent 6 found 14 changes (2 Critical) typed as New / Amended / Removed / Stricter / Relaxed, with a CRO-level impact summary.

5. **What-if Simulation**
   "RBI reduces KYC refresh for high-risk customers from 2 years to 6 months" → projected score 54% (−7), 1 new High gap, 3 required actions.

What the agents have to catch

Regulation · clause 1.2

"Fresh KYC … High Risk — **every 2 years**"

Bank policy · CTL-C

"High Risk customers **every 3 years**"

A control exists, but it is weaker than the obligation. Live run: the Mapper rated it **Partial (0.7)** and the Gap agent raised GAP-006.

Recorded live run · built-in sample · gpt-4.1

18 / 18
clauses extracted as obligations

9 gaps
2 Critical · 3 High · 3 Medium · 1 Low

61.1%
weighted score (9 Full · 4 Partial · 5 None)

~31 s
end to end, 6 LLM calls

SOLUTION COVERAGE

Feature Coverage vs. Hackathon Requirements

01	Regulatory knowledge ingestion	Full	02	Regulatory change intelligence	Full
03	Obligation extraction	Full	04	Control-framework understanding	Full
05	Regulatory-to-control mapping	Full	06	Control effectiveness assessment	Full
07	Evidence-based assessment	Partial	08	Gap identification	Full
09	Risk-based gap prioritization	Full	10	Remediation recommendation engine	Full
11	Autonomous regulatory impact analysis	Full	12	What-if / simulation capability	Full
13	Cross-regulation intelligence	Roadmap	14	Regulatory contradiction detection	Roadmap

Hackathon Threshold

F1 — ≥3 features	✓ Pass
F2 — ≥6 features	✓ Pass
F3 — ≥8 features	✓ Pass (11+)

Solution Depth

D1 — Textual, acceptable	✓ Exceeded
D2 — Textual, high reliability	✓ Claimed
D3 — Multimodal	Roadmap

Self-declared position

F3 / D2

11 of 14 features · structured-text input with built-in reliability controls

BUSINESS IMPACT

Business Value for Banking Institutions

Productivity · Faster first pass

1 run

replaces a document-by-document manual read for the first-pass gap analysis; analysts review findings instead of building them

Risk reduction · Earlier detection

Pre-audit

gaps and weak controls are surfaced and risk-ranked before an inspection finds them; change analysis runs the day an amendment lands

Fine avoidance · Better decisions

What-if

leadership sees the projected compliance impact of a proposed rule before it takes effect and budgets remediation early

Quantified savings to be measured in a pilot against the bank's current manual baseline.

Who Benefits

Compliance Teams
Automated first-pass analysis. Focus analyst time on judgment, not document hunting.

CRO / Board
Board-ready executive summaries. Real-time compliance score. Risk-prioritized gap register.

Internal Audit
Full traceability chain from regulation to control to evidence. Audit-ready output.

RegTech / Consultancies
White-label platform to accelerate client compliance engagements.

Target Market

Indian PSBs & Private Banks | NBFCs | RBI / SEBI regulated entities | Basel III compliant institutions

SCALABILITY & RESPONSIBLE AI

Built to Scale. Built to Be Trusted.

Scalability Architecture

- ✓ **Stateless FastAPI backend**: scales horizontally behind a load balancer
- ✓ **SSE streaming**: long pipelines run without HTTP timeouts
- ✓ **Async executor**: LLM calls run in a thread pool; the event loop stays responsive
- ✓ **Independent analyses**: no shared state between runs, so batch mode is a queue away
- ✓ **Long documents**: next step is chunked retrieval to replace today's 8,000-character input cap

Responsible AI & Trust

- ✓ **Full traceability**: every gap links to the exact obligation and control behind it
- ✓ **Confidence scoring**: a 0–1 confidence per mapping, shown in the traceability view
- ✓ **Conservative prompting**: agents prefer reporting a gap over force-fitting a match
- ✓ **Low temperature (0.1)**: OpenAI and Groq calls tuned for repeatable, schema-conformant output
- ✓ **Provider-switchable**: OpenAI, Groq or Anthropic Claude via one env var

Hallucination Mitigation

Every agent is prompted for an exact JSON schema. Recovery logic salvages complete items from truncated output instead of guessing. Every mapping carries a confidence score.

Privacy & Security

Documents are processed in memory and nothing is persisted server-side. Text is sent only to the configured LLM provider; self-hosted models are on the roadmap for on-premises use.

Human Oversight

AI generates findings; humans validate. Every gap includes AI reasoning so compliance officers can verify or override the assessment.

FUTURE ROADMAP

What Comes Next

Phase 1 · Now

- ✓ 6-agent core pipeline (working)
- ✓ Regulatory change intelligence
- ✓ What-if simulation
- ✓ SSE real-time streaming
- ✓ OpenAI / Groq / Anthropic providers

✓ Shipped

→

Phase 2 · Next 3 months

- ✓ Chunked retrieval for long documents (vector store)
- ✓ Multimodal input (image/table OCR)
- ✓ Cross-regulation intelligence (F13)
- ✓ Contradiction detection (F14)
- ✓ Evidence document ingestion (F7)

Next

→

Phase 3 · 3–6 months

- ✓ Continuous monitoring (webhook on regulatory feeds)
- ✓ Multi-bank / multi-jurisdiction tenancy
- ✓ Compliance posture history & trending
- ✓ Audit trail with digital signatures
- ✓ Regulator-facing report export

Planned

→

Phase 4 · 6–12 months

- ✓ Trainable on bank-specific policy corpus
- ✓ Fine-tuned compliance reasoning model
- ✓ API integration with core banking systems
- ✓ Real-time regulatory feed connectors (RBI, SEBI, BIS)
- ✓ Enterprise SaaS and self-hosted LLM option for on-prem banks

Planned

Path to F3 / D3

Phase 2 closes the three open features (F7 evidence, F13 cross-regulation, F14 contradictions) and adds multimodal input. Together with a repeat-run consistency benchmark, that is the evidence needed for a D3 claim.

SUMMARY

Claimed Position & Takeaway

9-Blocker Grid — Claimed Position



F3: 11 of 14 features fully covered (plus F7 partially), including simulation, change intelligence and risk-prioritized remediation.

D2: Structured textual input (PDF/DOCX/TXT), schema-constrained prompts, JSON recovery, per-mapping confidence and conservative "prefer a gap over a forced match" prompting.

regComply-AI

Regulations to Remediation. One agentic run.

ET AI Hackathon: Agentic Edition
Problem Statement 1 — Banking/Financial Regulations

F3/D2 | 7 Agents | 11 Features